

Vinicius F. da Silva

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Master's student in Computer Science at PPGCC (Graduate Program in Computer Science) at UFMG (Federal University of Minas Gerais), working at LaC (Compilers Lab), under the supervision of Dr. Fernando Quintão Pereira. Postgraduate student in Data Science specialization at UNICAMP (State University of Campinas).

AREAS OF INTEREST

- Compilers
- Code Optimization
- Formal Verification
- Virtualization and Containerization
- Operating Systems
- Programming Language Theory

EDUCATION

Federal University of Minas Gerais (UFMG) <i>MSc Student</i>	Belo Horizonte, Minas Gerais <i>9/2024</i>
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- **Research Title:** Automatic Generation of Benchmarks via Self-Similar Code
- **Supervisor:** Fernando Magno Quintão Pereira

Postgraduate Lato Sensu (UNICAMP) <i>Data Science</i>	Remote <i>8/2024</i>
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Federal University of Minas Gerais (UFMG) <i>Individual Course Enrollment</i>	Belo Horizonte, Minas Gerais <i>Completed</i>
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- **Subject:** Static Program Analysis

Pontifical Catholic University of Minas Gerais (PUCMINAS) <i>Bachelor's Degree in Computer Science</i>	Belo Horizonte, Minas Gerais <i>Completed</i>
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- **Undergraduate Final Project Title:** Maximizing Efficiency in Inter-Process Communication: Exploring a Zero-Copy Abstraction
- **Supervisor:** Pedro Henrique Penna

EXPERIENCE

Graduate Research at Compilers Lab (LaC) - DCC/UFMG	8/2024 – Current
Jr. Linux Kernel Developer <i>MagaluCloud</i>	6/2024 – 10/2024 <i>Remote</i>
Mid-Level FullStack Developer <i>Sociedade Mineira de Cultura</i>	3/2021 – 6/2024 <i>Belo Horizonte, Minas Gerais</i>
Intern Researcher Compilers and Operating System <i>Institute of Exact Sciences and Informatics - (ICEI) - (PUCMINAS)</i>	10/2020 - 12/2020 <i>Belo Horizonte, Minas Gerais</i>
PIBIC/CNPQ Researcher at Graduate Program in Dentistry <i>Institute of Biology Sciences - (ICBS) - (PUCMINAS)</i>	10/2019 - 8/2020 <i>Belo Horizonte, Minas Gerais</i>
Document: view	
Tutor - Compilers discipline <i>Institute of Exact Sciences and Informatics - (ICEI) - (PUCMINAS)</i>	4/2021 - 6/2021 <i>Belo Horizonte, Minas Gerais</i>
Document: view	
Tutor - Algorithms and Data Structures II discipline <i>Institute of Exact Sciences and Informatics - (ICEI) - (PUCMINAS)</i>	10/2021 - 01/2022 <i>Belo Horizonte, Minas Gerais</i>
Document: view	
Tutor - Data Base discipline <i>Institute of Exact Sciences and Informatics - (ICEI) - (PUCMINAS)</i>	7/2020 - 11/2020 <i>Belo Horizonte, Minas Gerais</i>
Document: view	

VOLUNTEERING

Artifact Evaluator <i>International Conference on Parallel Architecture and Compilation Techniques (PACT25)</i>	8/2025 - 9/2025 <i>Remote</i>
Artifact Evaluator <i>International Symposium on Code Generation and Optimization (CGO26)</i>	9/2025 - 12/2025 <i>Remote</i>
Artifact Evaluator <i>International Conference on Compiler Construction (CC26)</i>	12/2025 - 12/2025 <i>Remote</i>
Artifact Evaluator <i>Programming Language Design and Implementation (PLDI26)</i>	1/2026 - Current <i>Remote</i>

PUBLICATIONS

Vinícius Francisco da Silva, Leite, and Pereira 2025 paper
V. F. da Silva 2024 paper

REFERENCES

Silva, Vinícius Francisco da et al. (2025). “Multi-Language Benchmark Generation via L-Systems”. In: *arXiv preprint arXiv:2512.17616*.

Silva, V. F. da (2024). *Maximizando a Eficiência na Comunicação entre Processos: Explorando uma Abstração de Zero-Cópia*. Tech. rep. Belo Horizonte, Brazil: Pontifícia Universidade Católica de Minas Gerais (PUC Minas).

LETTER OF RECOMMENDATION

Pedro Henrique Penna

• [view](#)

Nanvix Operating System Internship

Huimin Cui and Gabriel Rodríguez

• [view](#)

PACT26 Artifact Evaluation Volunteership

Olivia Hsu

• [view](#)

CGO26 Artifact Evaluation Volunteership

SOFTWARE PATENTS

Software Patent - BR512022000810-6

• [view](#)

AnalisedoSorriso Mobile App

Software Patent - BR512022000377-5

• [view](#)

3DInteractiveDentalAnatomy Mobile App

Software Patent - BR512021000181-8

• [view](#)

InfoOdonto Mobile App

Software Patent - BR512020002866-7

• [view](#)

DTMOne Mobile App

Software Patent - BR512019000555-4

• [view](#)

OrthoHelp Mobile App